Explicit and Implicit Parallelism in R for Handling Big Data

Volume, velocity, variety and veracity are the four essential qualities of Big Data, and are the very reasons why it is challenging to process it using typical procedures. To process massive amounts of data quickly and efficiently, Big Data needs parallel computing. The R data analysis tool is useful for this.

It is quite difficult to efficiently deal with very huge data sets using current methodology and data mining software tools on a single personal computer. For Big Data, parallel computing platforms are seen to be a better answer. The idea of parallel computing is to break down a complex task into smaller chunks, each of which is handled by a single processor. These processes are carried out in a distributed and parallel fashion.

R is a widely used statistical software and data analysis tool written in an open source programming language. It is available on a variety of platforms, including Windows, Linux, and MacOS X. The R programming language is also an up-to-date tool. R has a number of parallel processing features and several packages that efficiently deal with parallel computing. When processing massive volumes of data, parallel programming can save a lot of time. Because modern computers have numerous processors and cores, as well as the capacity to perform hyper-threading, R has become compatible with them, allowing multiple simultaneous calculations on all resources.

Explicit parallelism in R

Explicit parallelism is defined by the inclusion of explicit expressions in the R programming language, which are used to describe (to a degree of detail) how parallel computing will be performed. R programming comprises several packages for explicit parallelism.

- **Rmpi**: In parallel computing, the alternative MPI (message passing interface) standard has recently become the *de facto* standard. This package in R makes it possible. There are many implementations of MPI. The popular ones are MPICH2, LAM-MPI and OpenMI.
- **pbdR**: This package is part of the 'Programming with Big Data in R' initiative. It's a distributed computing R tool that's highly scalable. MPI, ZeroMQ, ScaLAPACK, NetCDF4, PAPI, and other high-performance, high-level interfaces are included in this package.
- **snow**: This is the abbreviation for 'simple network of workstations'. Snow uses the master/slave communication model, in which one device or process (the master) commands one or more other devices or processes (known as slaves). It provides a high-level interface for parallel computations in R using a workstation cluster. It creates a virtual link between processes by implementing an interface to three different low-level mechanisms: socket, PVM (parallel virtual machine) and MPI (message passing interface).
- **foreach**: This enables parallel execution, which means it can run the same task on many processors/cores on your machine or multiple cluster nodes. It may be used with a variety of computational backbends including doParallel, doFuture, doRedis and doAzureParallel.
- **parSim**: Parallel simulation studies are what this stands for. It uses one or more computer cores to do flexible simulation studies. In addition to being able to run locally, the package is built to run on high-performance clusters.